

Book Review

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Machine Learning for Factor Investing: Python Version, by Guillaume Coqueret and Tony Guida. Boca Raton, Chapman and Hall–CRC. 2023. 358 pp., \$184.00 (hardbound), \$71.99 (e-book). ISBN 978-0-367-63974-7.

This book offers a clear and streamlined approach to applying machine learning models for factor investing, combining conceptual explanations with hands-on Python applications. While many existing resources on factor investing and quantitative finance focus on examples in R, this volume distinguishes itself by featuring Python-based code examples throughout, making it highly accessible to the growing community of finance and data science practitioners who prefer or require Python for their workflows.

The opening section establishes a foundation for applying machine learning to factor investing. It introduces clear notation, a rich dataset of firm-level predictors, and essential concepts in factor investing. The discussion covers asset pricing anomalies such as size, value, momentum, investment, and profitability, which motivate the strategies under study. The section emphasizes data preprocessing, feature engineering, and challenges of financial time series, including non-stationarity, noise, and overfitting, preparing readers for model development.

The second section surveys a comprehensive suite of supervised algorithms essential for modern factor investing, which covers Chapters 5–9. Penalized regressions like LASSO and elastic net demonstrate regularization for sparsity and robustness. Tree-based methods, including decision trees and random forests, capture nonlinear interactions. Neural networks model complex relationships, while support vector machines provide margin-based classification and regression. Bayesian methods offer probabilistic frameworks for uncertainty quantification. Each algorithm is paired with Python code, enabling hands-on experimentation.

The third section bridges the gap between modeling and practical investment implementation, encompassing Chapters 10 through 12. It covers model validation, hyperparameter tuning, cross-validation, and performance metrics, highlighting robust model selection. Ensemble models demonstrate how combining predictors enhances stability and accuracy. Portfolio backtesting translates model outputs into trading strategies, assessing performance while considering transaction costs and turnover. Practical Python workflows guide readers from algorithmic predictions to investment decisions.

The concluding chapters explore advanced methodologies. Interpretability techniques clarify how models translate inputs into outputs. Discussions on causality and non-stationarity help distinguish causal relationships from correlations and handle evolving financial patterns. Unsupervised learning uncovers inherent data structures for feature extraction and exploration. Reinforcement learning introduces dynamic decision-making in portfolio optimization, showing how agents learn optimal strategies through interaction with the environment.

Structured in a tutorial format, the book includes a wealth of applied examples and exercises that make the concepts accessible. It is also suitable for practitioners from other fields, such as medicine, who are new to Python. The authors first explain each topic at an intermediate technical level, then illustrate it with a Python implementation. However, the book covers only the basics of Python, and most algorithms rely on the `sklearn` library, without exploring more advanced

packages such as those in `scikit-learn-contrib`. Its focus is on practical application rather than model development, making it unsuitable as a companion for doctoral-level statistical courses.

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